1. ***This is the start of a multipart question. The information below pertains to question 1 to question 8.***

   The data set `fertil1` contains data from women in the US between the ages of 35 and 54. You can load the data into R using the `wooldridge` library using:

   ```
   library(wooldridge)
   dat = wooldridge::fertil1
   ```

   The main outcome of interest is the variable *kids*, which is the number of children the women has given birth to.
   What is the mean of the variable *kids* in the dataset?

   a. 2.7431
   b. 1.8002
   c. 3.8678
   d. 2.5798
   e. None of the above

2. How many missing values or NA's are there for the variable *age* in the dataset?

   a. 0
   b. 265
   c. 20
   d. 33
   e. None of the above

3. Consider the following linear regression model.

$$\begin{aligned} kids = \beta_0 + \beta_1(educ) + \beta_2(age) + \beta_3(age^2) + \beta_4(year) + \\ \beta_5(east) + \beta_6(northcen) + \beta_7(west) + \varepsilon \end{aligned}$$

   The model includes controls for education, age and age squared. The model also controls for *year*, which is year. Finally, (*east, northcen, west*), indicator variables for regional areas are included in the model.
   Estimate the model above with OLS. What is the estimated coefficient on education?

   a. –0.06932
   b. –0.13712
   c. –0.15261
   d. –0.17425
   e. –0.12260

4. What is the sum of squared errors of the model?

   a. 6632.205
   b. 2809.876

c. 5223.497
d. 861.390
e. None of the above

5. Suppose we wanted to create a confidence interval for one of the parameters of the model. If we wanted to create a 97.50% confidence interval, what critical value would we use?

   a. 2.4559
   b. 3.1180
   c. 2.2444
   d. 2.2664
   e. None of the above

6. Consider the null hypothesis that $H_0 : \beta_{northcen} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

   a. Reject the null at the 0.1% level
   b. Reject the null at the 1% level
   c. Reject the null at the 5% level
   d. Reject the null at the 10% level
   e. Fail to reject the null at the 10% level

7. Consider the null hypothesis that $H_0 : \beta_{educ} = -0.15$. What is the p-value associated with this hypothesis test?

   a. 0.1998
   b. 0.1639
   c. 0.1882
   d. 0.1510
   e. None of the above

8. What is the covariance between the estimate of the parameter on *(Intercept)* and the parameter on *west*?

   a. –0.03627
   b. –0.04409
   c. –0.04026
   d. –0.02994
   e. –0.02254

9. ***This is the start of a multipart question. The information below pertains to question 9 to question 15.***

   The data set `catholic` contains data on students that have recently left high school. You can load the data into R using the `wooldridge` library using:

```
library(wooldridge)
dat = wooldridge::catholic
```

The variable of interest is *hsgrad* which indicates if the student graduated high school. We would like to study what are the main determinants of high school graduation using a linear probability model. To answer this question you will need to add a new variable to the data set called *nonwhite* which should equal 1 if the student is non-white (i.e., asian, hispanic, or black) and zero otherwise.
What percent of the population are *nonwhite*?

a. 16.48%
b. 22.58%
c. 16.18%
d. 15.43%
e. None of the above

10. Estimate a linear probability model on high school graduation *hsgrad*.
Include in your model controls for if the student attended a Catholic high school (*cathhs*), *nonwhite*, and both the 12th grade reading and math scores. Your model should have 5 parameters including the intercept.
What is the R-squared from the estimated model?

a. 0.03834
b. 0.04439
c. 0.04080
d. 0.07493
e. 0.05675

11. What is the correct interpretation of the parameter on *cathhs*? Holding fixed the variables in the model, attending a Catholic high school increases the probability of high school graduation by . . .

a. 3.0198 percent
b. 0.3020 percent
c. 3.0198 percentage points
d. 0.0302 percent
e. None of the above

12. What is the estimate of the lower bound of the 90% confidence interval for the intercept of the model?

a. 0.3342
b. 0.2780
c. 0.4815
d. 0.5120
e. None of the above

13. Consider the null hypothesis that $H_0 : \beta_{math12} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

    a. Reject the null at the 0.1% level
    b. Reject the null at the 1% level
    c. Reject the null at the 5% level
    d. Reject the null at the 10% level
    e. Fail to reject the null at the 10% level

14. Re-estimate the model adding log family income (*lfaminc*) as an additional control. What is the percentage change in the parameter on *cathhs* when we control for family income?

    a. –53.62%
    b. –51.42%
    c. 47.80%
    d. –54.01%
    e. None of the above

15. Re-estimate the model adding log family income (*lfaminc*) as an additional control. What is the variance inflation factor for the parameter on *cathhs* when we control for family income? (note: if you are calculating the variance inflation factor from the formula you need to ensure that the same observations are used in the different regressions)

    a. 1.0249
    b. 1.2479
    c. 1.2946
    d. 0.9104
    e. None of the above

Solution Key

Q	Ans	Value	How / trap
1	**a**	2.7431	`mean(dat$kids)`
2	**a**	0	`sum(is.na(dat$age))`
3	**e**	–0.12260	`lm(kids ~ educ + age + I(age^2) + year + east + northcen + west)`
4	**b**	2809.876	SSR = `sum(resid(m)^2)`
5	**c**	2.2444	`qt(1-0.025/2, df.residual(m))`, df = 1129 – 8 = 1121
6	**c**	$p = 0.0237$	$0.01 < p < 0.05 \Rightarrow$ reject at 5%
7	**e**	0.1351	$t = (\hat{\beta} + 0.15)/se$, two-sided `2*pt(-|t|, df)`
8	**d**	–0.02994	`vcov(m)["(Intercept)","west"]`
9	**b**	22.58%	`nonwhite = asian|hispan|black`, then `mean`
10	**d**	0.07493	$n = 5970$; *hsgrad* has 1460 NAs (lm drops them)
11	**c**	0.0302	LPM coefficient $\times 100$ = percentage points
12	**d**	0.5120	`confint(m1, level=0.90)`
13	**a**	$p \approx 10^{-43}$	$p < 0.001$
14	**e**	–39.46%	`100*(b_new/b_old - 1)`
15	**a**	1.0249	`car::vif(m2)`; by hand only on rows with non-NA *hsgrad* (full sample gives 1.0167)

R code

```
# ECON 5021 Unit 2 Data Analysis Practice: 26101200001 - solution key
# Key: 1a 2a 3e 4b 5c 6c 7e 8d 9b 10d 11c 12d 13a 14e 15a

library(wooldridge)
library(car)

# ---- Q1-8: fertil1 ----
dat = wooldridge::fertil1

# Q1. What is the mean of the variable kids in the dataset?
mean(dat$kids)                                  # 2.7431  (a)

# Q2. How many missing values or NA's are there for the variable age?
sum(is.na(dat$age))                             # 0       (a)

# Q3. Estimate the model with OLS. Coefficient on education?
m = lm(kids ~ educ + age + I(age^2) + year + east + northcen + west, data = dat)
summary(m)                                      # -0.12260 (e)

# Q4. What is the sum of squared errors of the model?
sum(resid(m)^2)                                 # 2809.876 (b)

# Q5. Critical value for a 97.50% confidence interval?
```

```
qt(1 - 0.025/2, df = df.residual(m))             # 2.2444  (c)   df = 1121

# Q6. H0: b_northcen = 0. Strongest conclusion?
summary(m)$coefficients["northcen", ]            # p = 0.0237 -> reject at 5% (c)

# Q7. H0: b_educ = -0.15. p-value?
t = (coef(m)["educ"] - (-0.15)) / sqrt(vcov(m)["educ", "educ"])
2 * pt(-abs(t), df = df.residual(m))             # 0.1351  (e)

# Q8. Covariance between the estimates on (Intercept) and west?
vcov(m)["(Intercept)", "west"]                   # -0.02994 (d)

# ---- Q9-15: catholic ----
dat = wooldridge::catholic
dat$nonwhite = as.numeric(dat$asian == 1 | dat$hispan == 1 | dat$black == 1)

# Q9. What percent of the population are nonwhite?
100 * mean(dat$nonwhite)                         # 22.58%  (b)

# Q10. LPM of hsgrad on cathhs, nonwhite, read12, math12. R-squared?
m1 = lm(hsgrad ~ cathhs + nonwhite + read12 + math12, data = dat)
summary(m1)$r.squared                            # 0.07493 (d)   n = 5970 (hsgrad has 1460 NAs)

# Q11. Interpretation of the parameter on cathhs?
coef(m1)["cathhs"]                               # 0.030198 -> 3.0198 percentage points (c)

# Q12. Lower bound of the 90% CI for the intercept?
confint(m1, level = 0.90)                        # 0.5120  (d)

# Q13. H0: b_math12 = 0. Strongest conclusion?
summary(m1)$coefficients["math12", ]             # p ~ 1e-43 -> reject at 0.1% (a)

# Q14. Add lfaminc. Percentage change in the parameter on cathhs?
m2 = lm(hsgrad ~ cathhs + nonwhite + read12 + math12 + lfaminc, data = dat)
100 * (coef(m2)["cathhs"] / coef(m1)["cathhs"] - 1)   # -39.46% (e)

# Q15. VIF for cathhs when we control for lfaminc?
vif(m2)["cathhs"]                                # 1.0249  (a)
# by hand: same observations as m2 (drop the rows where hsgrad is NA)
d = dat[!is.na(dat$hsgrad), ]
1 / (1 - summary(lm(cathhs ~ nonwhite + read12 + math12 + lfaminc, data = d))$r.squared)  #
    1.0249
```